

Milk Thistle & Liver Support

A practical reference for using silymarin (milk thistle extract) to support liver enzyme health, based on the current clinical evidence — not folklore.

What Silymarin Actually Does

Silymarin is the flavonolignan complex extracted from milk thistle seeds, and it is the compound responsible for the plant's liver effects. It acts as an antioxidant, helps stabilize liver cell membranes, and may support glutathione regeneration. Intravenous silibinin — a purified silymarin component — is the established clinical antidote for death cap mushroom poisoning in European hospitals.

What the Evidence Supports

- **NAFLD / elevated liver enzymes**
Meta-analyses of randomized trials show consistent, statistically significant reductions in ALT and AST with silymarin vs. placebo.
- **NASH (steatohepatitis)**
High-dose silymarin (700mg/day) produced measurable enzyme and some histological improvement over 48 weeks in the SynCH clinical trial.
- **Alcohol-related liver disease**
Cochrane review found improved enzyme markers in several trials, but insufficient evidence for a mortality benefit.
- **What it will not do**
Reverse cirrhosis, replace reducing alcohol intake, or substitute for medical evaluation of abnormal liver enzymes.

Dosing & Quality Checklist

Standard Dosing Protocol

- **Standard maintenance dose**
140mg standardized silymarin extract (70–80% silymarin content), 2–3x daily. Total: 280–420mg/day.
- **Higher clinical dose**
700mg/day (split doses) — the dose used in the strongest NASH trial data. Discuss with a physician before exceeding standard dosing.
- **Timing**
Take with meals to reduce mild GI upset; effects on enzyme markers typically assessed after 8–12 weeks of consistent use, not days.

Quality Checklist – Look For

- **Standardized extract**
Label states silymarin content (70–80%) — not just 'milk thistle powder' by raw weight, which varies widely in actual potency.
- **Bioavailability carrier**
Phosphatidylcholine-bound formulations ("phytosome"/"Siliphos") absorb significantly better than plain extract.
- **Third-party testing**
USP, NSF, or independent lab verification — silymarin content has been shown to vary significantly between untested commercial products.
- **Allergy caution**
Avoid if allergic to ragweed, chrysanthemums, or marigolds — milk thistle is a botanical relative and cross-reactivity is documented.

When to See a Doctor Instead

Persistently elevated liver enzymes, jaundice, unexplained fatigue with abdominal discomfort, or any known liver condition should be evaluated medically before self-treating with any supplement, including milk thistle.